

# Course Visualization Tool – Final Project Documentation

## CIT 312 – Advanced Web Design

Jasly Delmonte

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### 1. Project Overview

The Course Visualization Tool is an interactive web application designed to showcase the nine undergraduate programs within Purdue University's School of Applied & Creative Computing. The project uses Bootstrap, SCSS, semantic HTML, and JavaScript to create a professional, responsive, and accessible experience.

#### The application includes:

- A card-based layout displaying all 9 programs
- A modal for each program listing required courses and career paths
- A 10-question interactive quiz recommending programs
- A responsive hamburger menu
- A custom favicon
- Fully responsive layout
- Accessibility-friendly structure

### 2. Use of SCSS (SASS Preprocessor)

SCSS was used to organize and enhance the styling of the application.

#### SCSS Features Used

- **Variables** Used to store Purdue-branded colors for consistent styling.
- **Nesting** Applied to structure selectors cleanly and logically.
- **Color Functions** `darken()` used to create subtle hover effects.
- **Compilation Workflow** SCSS was compiled into `main.css` using the Live Sass Compiler in VS Code.

#### Purpose

Using SCSS allowed for:

- Cleaner, more maintainable code
- Consistent color usage
- Easier responsive adjustments
- Professional-level styling

### 3. Responsiveness

Responsiveness was achieved through a combination of Bootstrap's layout system and custom SCSS.

#### Bootstrap Grid System

- .container, .row, .col
- Responsive column classes such as row-cols-1 and row-cols-md-3
- Utility spacing classes like my-5, g-4, and p-4

#### Media Queries

Custom SCSS media queries adjust:

- Card image heights on smaller screens
- Navigation width on larger screens

#### Mobile Navigation

The hamburger menu collapses into a slide-in panel on small screens and displays as a static horizontal menu on larger screens.

### 4. Hamburger Menu

The navigation menu was built using a CSS-only slide-down technique.

#### Key Features

- Font Awesome icons for open/close
- nav:target used to reveal the menu
- CSS transforms for smooth sliding
- No JavaScript required
- Fully responsive

#### Why This Matters

This demonstrates understanding of:

- CSS transforms
- Target selectors
- Mobile-first navigation patterns

### 5. Favicon

A custom favicon was included using:

html

```
<link rel="icon" type="image/png" href="images/favicon.png">
```

#### Requirements Met

- Original favicon
- Correct file type
- Correct placement inside <head>
- Matches project theme

## 6. Bootstrap Features Used

The project uses a wide range of Bootstrap components and utilities, including:

### Components

- Cards
- Modals
- Buttons
- Scrollable modal dialogs

### Utilities

- Spacing (p-4, my-5, g-4)
- Borders (border, rounded)
- Typography (display-4, lead, text-center)
- Layout (container, row, col)

### Why Bootstrap Was Effective

Bootstrap provided:

- A clean card layout
- Professional modals
- Consistent spacing
- Fast, reliable responsive design

## 7. Accessibility

Accessibility was integrated through:

### Semantic HTML

- `<header>`, `<nav>`, `<main>`, `<section>`, `<footer>`

### ARIA Attributes

- `aria-labelledby`
- `aria-hidden="true"`
- `tabindex="-1"`

### Alt Text

All program images include descriptive alt attributes.

### Keyboard-Friendly Buttons

All quiz and modal buttons are real `<button>` elements.

## 8. Interactive Quiz

A 10-question quiz was implemented using JavaScript to create an interactive, dynamic experience.

### Features

- Uses arrays, functions, and event listeners
- Displays one question at a time

- Recommends programs based on user responses
- Includes a restart button
- Fully dynamic without reloading the page